Thresholds and Learning in Adaptive Intrusion Detection: A Two-Capture Case Study

Oleksandr Kuznetsov[1,2,4,*] Emanuele Pio De Bernardis[1]
Emanuele Frontoni[3]

[1]Department of Theoretical and Applied Sciences, eCampus University,
Via Isimbardi 10, 22060 Novedrate (CO), Italy

[2]SMARTEST Research Center, eCampus University, Novedrate, Italy

[3]Department of Political Sciences, Communication and International Relations,
University of Macerata, Via Crescimbeni, 30/32, 62100 Macerata, Italy

[4]Department of Intelligent Software Systems and Technologies,
School of Computer Science and Artificial Intelligence,
V. N. Karazin Kharkiv National University, 4 Svobody Sq., 61022 Kharkiv, Ukraine

*Correspondence: oleksandr.kuznetsov@uniecampus.it

E. Frontoni: emanuele.frontoni@unimc.it

ORCID: 0000-0003-2331-6326 (O.K.); 0009-0004-7915-5204 (E.P.D.B.);
0000-0002-8893-9244 (E.F.)

1 November 2026

Abstract

Threshold selection and calibration-label eligibility can change conclusions about adaptive intrusion detection even when label acquisition is fixed. We develop an auditable evaluation procedure and apply it to two chronological network captures under sparse, delayed supervision. Controls separate score refitting, calibration-memory capacity, and threshold objectives. On TON_IoT, refitting an additive spline head with the same CAL64 observations and update times as its frozen control lowers false-positive rate (FPR) by 28.15 percentage points while losing 1.91 points of recall at the balanced-accuracy objective. However, a stronger all-label threshold-only policy reaches 86.78% balanced accuracy after changing its cost weight, exceeding the default head's 84.84%. At a nominal 20% calibration FPR target, the head attains 91.50% recall versus 80.53%, with achieved FPRs of 19.95% and 18.37%; these are nearby, not identical, operating points. At 13% FPR, paired interpolation gives a recall difference of 1.49 ± 3.66 points. The tested head controllers miss the stricter 5% and 10% targets even with larger calibration memory. On UNSW-NB15, historical frozen policies outperform the tested adaptive alternatives by balanced accuracy. Baseline audits also expose poor transfer and sensitivity to forest training order. Five seeds describe variation conditional on each capture; head configuration sensitivity on TON spans 81.49–88.97% balanced accuracy, substantially wider than the default within-configuration seed SD of 0.79 points. These exploratory case studies provide a reproducible procedure for attributing error trade-offs, rather than a new detection architecture or evidence of uniform adaptive superiority.

Keywords: intrusion detection; delayed labels; threshold selection; error trade-offs; temporal evaluation; reproducibility.

1 Introduction

An intrusion detector assigns a score to each observation and compares it with a decision threshold. Updating the score and moving the threshold are different interventions. Either can change the

number of missed attacks and false alarms. A comparison of adaptive and static detectors therefore needs controls that identify which score, threshold, and calibration-data interventions supply the reported improvement.

This question becomes more difficult when labels are scarce and delayed. The observations available for refitting may differ from those available for threshold selection. A small calibration memory can favor a different operating point from a larger memory, even when both optimize the same objective. Furthermore, a frozen model can be a weak reference because its initial fit transfers poorly to the evaluation period. These effects can make a large adaptive-versus-frozen contrast misleading as evidence of general learning superiority.

We study these issues using chronological TON_IoT and UNSW-NB15 replays. The principal question is: *what improvement remains after specifying a strong threshold-only controller, a relevant operating point, and a clear label-use contract?* The model families are an additive spline detector, logistic regression (LR), and an adaptive random forest (ARF). The spline intervention refits only eight feature gains and an intercept, providing a compact, explicit change to the score function.

The evidence has a defined scope. The historical TON campaign supplies an archived four-trace experiment. Subsequent controls and the present diagnostics were motivated by its observed outcomes. Both extensions are exploratory on TON. UNSW is a separate controlled capture, but had appeared elsewhere in the project. Neither capture is an untouched prospective field test. All reported seed variation is conditional on one stream per dataset.

The contribution is a reproducible evaluation procedure and a case study containing both positive and negative results: paired score and memory interventions, causal threshold selection, explicit label-use contracts, baseline-quality audits, and representation recurrence analysis. The results below emphasize operating policies and baseline quality. The Supplement retains the full historical campaign, implementation details, attack-family results, resource accounting, and sensitivity tables.

2 Related work

Concept-drift research distinguishes changes in a data-generating process from the mechanisms used to detect or respond to them [1]. Prequential evaluation requires prediction before an observation is used for learning [2]; delayed-label evaluation makes the feedback schedule explicit [3]. Adaptive random forests combine online ensembles with drift-triggered replacement [4]. Active learning addresses the separate question of which labels to request [5]. We use uniform requests so that acquisition remains fixed across controllers.

In security, INSOMNIA combines adaptation with active learning and label estimation [6]; continuous Android malware learning similarly couples representation updates and selective labeling [7]. Transcend and CADE study drift detection and explanation [8, 9]. Our focus is the downstream attribution of decision changes once feedback is available. Temporal and spatial biases, deployment assumptions, and dataset construction remain central to this interpretation [10–13].

Threshold selection differs from probability calibration [14]. Here CAL denotes examples used to select a binary threshold, without asserting that scores are posterior probabilities. Prior-shift correction provides an intercept-based reference [15, 16], while changing class-conditional traffic can require more than an intercept. The spline score is a logistic generalized additive model [17]; its fixed univariate representation also relates to neural additive models and KANs [18, 19]. The study evaluates update policies rather than proposing a new architecture.

Table 1: Evaluation streams. Observed prevalence is retained. Each dataset supplies one recorded stream, shared by all five algorithmic seeds.

Stream	D records	Normal	Attacks	Hours
TON_IoT	3,549,973	309,673	3,240,300	62.77
UNSW-NB15	1,016,018	803,736	212,282	7.89

3 Data and causal evaluation

3.1 Captures, partitions, and feature availability

TON_IoT supplies network records from a controlled IoT/IIoT environment [20, 21]. We retain eight complete-flow numerical features: duration, source/destination bytes, missed bytes, and source/destination packet and IP-byte counts. Records are ordered by start time plus duration. A fits the initial transformation and model, B selects historical thresholds, C supports the original development, and D is evaluated. The full dates, preprocessing counts, and earlier structural exposure are recorded in Supplement S1.

UNSW-NB15 supplies a separately collected laboratory capture [22, 23]. We use its four full network CSVs, remove only three verified duplicate file-boundary rows, and retain 2,540,044 records. The eight numerical fields are `dur`, `sbytes`, `dbytes`, `sloss`, `dloss`, `Spkts`, `Dpkts`, and `tcprtt`. This is a compact feature contract with different semantics from TON. Records are sorted by $\max(\text{Ltime}, \text{Stime} + \text{float32}(\text{dur}))$, then stable row identity, and split chronologically 40%/20%/40% into A/B/D. Equal timestamps at boundaries are retained without an embargo. Models are fitted anew on each dataset.

3.2 Models, feedback, and memory

All models use a fixed $\log(1 + \max(x, 0))$ transformation and standardization fitted on A. The initial LR and spline fits use the same seeded 400,000-row sample of A. LR uses the eight transformed inputs. The spline model fits eight cubic B-spline basis coefficients per feature, with subsequent adaptation restricted to

$$s(x; g, b) = b + \sum_{j=1}^{8} g_j f_j(x_j), \tag{1}$$

where the eight functions f_j remain fixed. A head update fits eight gains and an intercept by regularized logistic regression on the current FIT memory. The original refit uses zero-centred L2 regularization; a previous-head-centred alternative is retained in the Supplement.

The stream is predicted in 10,000-row blocks. Uniform requests select 1% of each block without replacement. Requested labels from block k arrive after prediction of block $k + 1$ and first affect block $k+2$. The last incomplete block is retained. All methods use the same requested row identities within a dataset, seed, delay, and budget condition. The current sampler uses `SeedSequence([seed, block, 20261101])`. The archived campaign's older sampler is used only in its separate numerical bridge.

A salted hash of each raw feature vector assigns it permanently to FIT or CAL with nominal probabilities 3/4 and 1/4. The default memories hold 192 FIT and 64 CAL observations, with equal class quotas for eviction; one class can occupy the other class's unused slots. Repeated observations can occupy separate positions. CAL entries are eligible for 50 blocks from observation time; at least five observations of each class are required. FIT has no age filter. A head and its matched threshold change atomically only when the transaction is valid. A separate CAL256 memory accepts all requested labels and updates a frozen detector's threshold independently. We

Table 2: Information contracts. All rows share acquisition and feedback timing. Model state and pending labels are additional to the listed example memories.

Controller	Learning	Threshold selection
Frozen / B	A only	Historical B
Frozen / CAL256	A only	All requested labels; recent CAL256
Head / CAL64	Disjoint FIT192	Recent CAL64; valid head transaction
Frozen/head / role256	A only or fixed FIT192 trajectory	Same CAL-role256 and matched transaction times
ARF / CAL64 or CAL256	Every arriving requested label	CAL labels reused in learning
ARF FIT-only / CAL64	Arriving FIT-role labels only	Disjoint CAL64

distinguish this *all-label CAL256* from the disjoint *CAL-role256* used in the matched-memory controls below. Equal capacity alone does not imply equal calibration observations.

ARF uses River 0.22.0, ten trees, and defaults apart from the seed [24]. The first extension processes the A sample chronologically. The present audit additionally fits the identical sample in its seeded random order. Both initial states are evaluated frozen and with feedback. The FIT-only variant starts from the chronological state and excludes every CAL-role label from learning. This aligns label disjointness with the head experiment, while retaining the forest's persistent tree memory. It does not impose the finite FIT192 retention contract. Thus ARF is an equal-acquisition-budget comparator with explicitly different model state and, for its standard variant, different label reuse.

3.3 Controls with identical calibration observations

The original Frozen/CAL256 versus Head/CAL64 contrast changes score learning, calibration capacity, label eligibility, and the threshold-update gate. It estimates a complete policy-package difference. To isolate the score intervention, we cross frozen and adapted scores with CAL-role capacities 64 and 256 on both streams and all five seeds. Within each capacity, the two scores use exactly the same labeled occurrences, age filter, objective, tie rule, and update times. Both CAL buffers remain disjoint from FIT192 and use the default role salt.

The head follows the original FIT192/CAL64 trajectory in every cell. All four thresholds change only at successful transactions of that trajectory. Thus increasing CAL capacity changes threshold evidence while holding the fitted score sequence and its gate fixed. CAL-role256 retains up to 448 FIT+CAL observations in total, compared with 256 for FIT192/CAL64, without extra label requests. It is a controlled memory intervention, rather than a fully retuned 448-example learner. Every shared per-block prediction count and threshold agrees exactly with the retained earlier replay.

3.4 Threshold objectives and matched-error descriptions

A decision is positive when $s(x) > \tau$. The historical and recent thresholds minimize

$$L_\lambda(\tau) = \mathrm{FNR}(\tau) + \lambda\,\mathrm{FPR}(\tau). \tag{2}$$

The first extension fixes $\lambda \in \{0.25, 0.5, 1, 2, 4\}$; the current audit adds 0.75, 1.25, 1.5, and 3. Every threshold is fitted using past available labels. At $\lambda = 1$, this is equivalent to maximizing balanced accuracy (BA). Exact candidate and tie rules appear in Supplement S1. For attack prevalence π and costs $C_{\mathrm{FN}}, C_{\mathrm{FP}}$, the rate weight is $C_{\mathrm{FP}}(1-\pi)/(C_{\mathrm{FN}}\pi)$. A value of one is a declared evaluation objective, rather than a universal operational cost.

Two complementary checks address operating-point comparability. First, we interpolate the observed FPR/recall polylines within their support, separately for each paired seed. This is a retrospective description of recorded policies. It neither selects a deployable threshold nor estimates an oracle ROC curve. Interpolating a curve of seed means is also reported and kept distinct from averaging the paired-seed comparisons.

Second, causal FPR-target controllers use only past normal calibration scores. For target $\alpha \in \{0.05, 0.10, 0.20\}$, they select the smallest observed negative-score threshold with empirical FPR at most α, respecting complete ties. B supplies the initial threshold. The original package comparison uses all-label CAL256 for the frozen score and disjoint CAL64 for the head. The matched-memory audit also evaluates both scores on identical CAL-role64 and CAL-role256. Requested targets are identical; achieved future FPR can differ under distribution change. We report both achieved FPR and recall rather than assuming that a calibration target is a future guarantee.

3.5 Metrics, overlap, and evidence status

We sum confusion counts over D before calculating recall, FPR, BA, and precision. AUROC and noninterpolated average precision (AP) are also calculated within prediction blocks containing both classes. We report macro, normal-count-weighted, and attack-count-weighted summaries and their coverage. Pooled AUROC/AP are used only for a fixed score. Pooling adaptive scores would compare values from different fitted states. AP is the recall-increment-weighted mean of precision, as implemented by scikit-learn, rather than a trapezoidal interpolation [25, 26].

Seen/unseen strata refer to exact equality of eight stored float32 features against earlier A/B/C vectors on TON and A/B vectors on UNSW. They describe representation recurrence, not event identity, attack-family novelty, or memorization. Both strata are evaluated on the same adaptive trajectory.

Seeds 42–46 vary the initial sample and label requests. All seed means and sample SDs are *conditional on the same recorded stream*; five seeds are not five independent captures. The historical acceptance rule required joint non-increase of FN and FP, with at least one strict improvement, in four of five spline seeds against Frozen/CAL64. It was a versioned empirical rule, not an externally registered trial or significance test. Analogous counts for other models and controls are exploratory. Original results, first-extension results, and new diagnostics are retained separately.

3.6 Computational assistance and verification

OpenAI ChatGPT/Codex was used under the corresponding author's direction to assist with diagnostic protocol formulation, implementation, execution of replay scripts, numerical analysis, and plotting (November 2026 sessions). The exact backend model snapshot was not retained. Reproducibility is based on the archived executable source, fitted initial states, pinned numerical environment, input digests, per-block reports, and deterministic analysis scripts. Figures are generated from recorded numerical outputs with Matplotlib. The matched-memory audit verifies 274,090 replay invariants and exact shared-trajectory comparisons, followed by 1,930 source and aggregation checks. AI assistance is a tool-use disclosure; responsibility for scientific interpretation and the submitted work rests with the human authors.

Table 3: Matched calibration observations and update times. CAL is the capacity of the disjoint CAL-role memory. The FIT192 head trajectory is held fixed. Rates are percentages, mean ± conditional seed SD; λ is the FP weight.

			Frozen score			Head score		
Stream	CAL	λ	Recall	FPR	BA	Recall	FPR	BA
TON	64	1	91.37 ± 2.24	47.93 ± 4.68	71.72 ± 1.73	89.46 ± 0.89	19.78 ± 0.77	84.84 ± 0.79
TON	64	2	86.40 ± 3.43	33.97 ± 2.30	76.22 ± 1.60	84.66 ± 0.94	9.57 ± 1.56	87.55 ± 0.98
TON	256	1	91.28 ± 2.84	45.31 ± 5.20	72.99 ± 1.34	89.72 ± 0.84	16.37 ± 0.96	86.67 ± 0.48
TON	256	2	86.41 ± 3.92	32.80 ± 2.45	76.80 ± 1.99	84.38 ± 0.77	6.59 ± 1.16	88.89 ± 0.89
UNSW	64	1	96.63 ± 0.25	9.12 ± 0.72	93.75 ± 0.34	96.79 ± 0.14	10.36 ± 0.50	93.22 ± 0.22
UNSW	64	2	96.25 ± 0.33	8.30 ± 0.40	93.97 ± 0.32	96.50 ± 0.19	9.53 ± 0.45	93.49 ± 0.18
UNSW	256	1	95.93 ± 0.13	8.91 ± 0.86	93.51 ± 0.40	96.60 ± 0.16	8.99 ± 0.53	93.81 ± 0.21
UNSW	256	2	95.22 ± 0.26	7.53 ± 0.60	93.84 ± 0.23	95.84 ± 0.49	7.78 ± 0.45	94.03 ± 0.29

4 Results

4.1 Score changes persist when calibration memory is matched

Table 3 separates score changes from CAL capacity. At $\lambda = 1$ on TON, the head-minus-frozen difference with identical CAL64 is -28.15 ± 5.22 percentage points in FPR and -1.91 ± 2.33 in recall. With identical CAL-role256, the corresponding differences are -28.94 ± 4.89 and -1.56 ± 2.37. A score intervention therefore changes the error compromise under this fixed role and calibration policy; unequal memory is not the sole explanation of the original contrast.

Increasing head calibration capacity from 64 to 256, while retaining the same head trajectory, adds 1.83 ± 0.51 BA points at $\lambda = 1$ on TON and 0.59 ± 0.22 on UNSW. On UNSW, however, the score contrast itself changes from -0.54 ± 0.39 BA points at CAL64 to $+0.30 \pm 0.24$ at CAL-role256, and both remain below historical Frozen/B. This interaction cautions against describing a head-learning benefit without its calibration contract.

The label-eligibility contract matters beyond capacity. On TON at $\lambda = 2$, frozen CAL-role256 yields 32.80% FPR and 76.80% BA, whereas all-label CAL256 yields 13.16% FPR and 86.78% BA. These memories have the same nominal capacity but different eligible observations and update gates. The larger role-restricted memory does not recover the all-label controller's behavior. A deterministic feature-role partition can also exclude recurring vectors from calibration altogether; this is a representativeness concern that increasing capacity alone cannot address. The experiments measure the policy contrast, without assigning all of it to one mechanism. Supplement S6 provides the complete same-memory target controllers, operating curves, supported matched-error descriptions, and per-seed counts.

4.2 The apparent head benefit depends on threshold policy

Table 4 compares the original complete policy packages against the stronger all-label threshold-only control. On TON at $\lambda = 1$, Head/CAL64 has 28.36 percentage points lower FPR and 2.92 points lower recall than Frozen/CAL256. Each seed has fewer FP and more FN. This is a difference between two complete update policies at one declared calibration objective.

Changing only the frozen detector's cost weight to $\lambda = 2$ raises its BA from 72.12% to 86.78%, above the head's 84.84% at $\lambda = 1$. At $\lambda = 2$, the head reaches 87.55% BA. The large 28-point FPR contrast therefore describes one objective and information contract, rather than a comparable advantage across operating points. The intermediate frozen weight $\lambda = 1.25$ gives 87.22% BA, illustrating the sensitivity to this choice without establishing that a weight selected after inspecting D will transfer. The observed operating curves in Figure 1 show where the gap changes.

Table 4: Spline operating policies. Rates are percentages, mean ± sample SD across five randomizations conditional on each stream. Each threshold uses only past labels.

Stream	Policy	Recall	FPR	BA
TON	Frozen / B	20.93 ± 0.37	12.59 ± 0.14	54.17 ± 0.18
TON	Frozen / CAL256, $\lambda = 1$	92.38 ± 1.32	48.14 ± 4.67	72.12 ± 2.42
TON	Frozen / CAL256, $\lambda = 2$	86.72 ± 1.73	13.16 ± 0.87	86.78 ± 0.93
TON	Head / CAL64, $\lambda = 1$	89.46 ± 0.89	19.78 ± 0.77	84.84 ± 0.79
TON	Head / CAL64, $\lambda = 2$	84.66 ± 0.94	9.57 ± 1.56	87.55 ± 0.98
UNSW	Frozen / B	94.53 ± 0.11	1.95 ± 0.02	96.29 ± 0.05
UNSW	Frozen / CAL256, $\lambda = 1$	96.37 ± 0.07	5.53 ± 0.12	95.42 ± 0.06
UNSW	Frozen / CAL256, $\lambda = 2$	96.02 ± 0.14	5.07 ± 0.15	95.47 ± 0.10
UNSW	Head / CAL64, $\lambda = 1$	96.79 ± 0.14	10.36 ± 0.50	93.22 ± 0.22
UNSW	Head / CAL64, $\lambda = 2$	96.50 ± 0.19	9.53 ± 0.45	93.49 ± 0.18

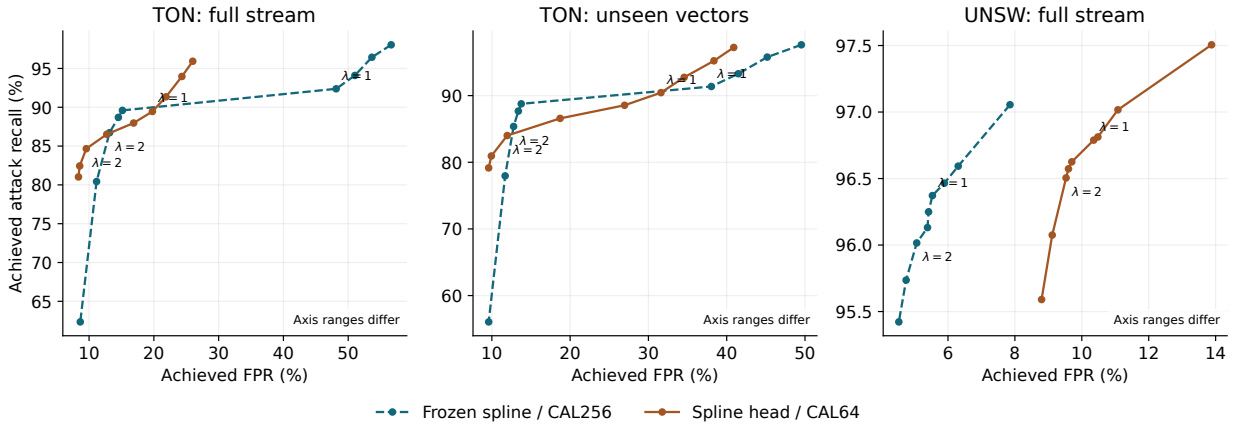


Figure 1: Causal threshold policies for the fixed cost grid. Points are means over five paired randomizations on each single stream. Lines connect implemented policies and do not constitute an oracle frontier. Axis ranges differ. The annotated weights illustrate why a single $\lambda = 1$ contrast cannot establish curve-wide superiority.

The original five-point grid also makes interpolation resolution important. In TON seed 42, the frozen thresholds at $\lambda = 0.25$ and 0.5 coincide in 238 of 355 blocks; those at 2 and 4 coincide in 201. Finite calibration data produce shared thresholds and coarse operating points. Supplement S4 provides each seed and distinguishes interpolation of the mean curve from paired-seed interpolation. The latter retains variation in the location of the operating points; the two summaries need not agree.

With the denser nine-weight grid, the paired head-minus-frozen recall difference at 13% FPR is 1.49 ± 3.66 percentage points (5 supported pairs). This remains a retrospective interpolation within each seed, rather than a threshold selected for a new stream.

On UNSW, the historical Frozen/B spline is strongest by mean BA among the default spline policies: 96.29%. Recent CAL64 selection lowers BA to 93.75%, CAL256 to 95.42%, and head/CAL64 to 93.22%. Thus neither tested recent threshold adaptation nor head refitting improves BA over the historical policy on this capture. A higher recall under another policy remains possible at a larger false-alarm cost. This qualifies any claim that adaptation generally fails or succeeds independently of the chosen utility.

Table 5: Requested calibration FPR versus achieved future rates for the spline score. Target and rates are percentages. Mean ± sample SD describes five randomizations of the same stream. Thresholds use only previously available normal labels.

Stream	Target	Controller	Recall	FPR	BA
TON	5	Frozen/CAL256	30.37 ± 4.04	4.57 ± 0.35	62.90 ± 2.04
TON	5	Head/CAL64	68.02 ± 8.09	12.07 ± 2.34	77.97 ± 3.26
TON	10	Frozen/CAL256	47.99 ± 8.74	7.02 ± 0.74	70.48 ± 4.04
TON	10	Head/CAL64	81.31 ± 6.49	16.03 ± 2.15	82.64 ± 2.62
TON	20	Frozen/CAL256	80.53 ± 6.16	18.37 ± 1.34	81.08 ± 2.98
TON	20	Head/CAL64	91.50 ± 1.01	19.95 ± 1.52	85.77 ± 1.16
UNSW	5	Frozen/CAL256	94.55 ± 0.80	6.20 ± 0.30	94.17 ± 0.29
UNSW	5	Head/CAL64	89.86 ± 3.66	9.21 ± 1.09	90.32 ± 1.62
UNSW	10	Frozen/CAL256	96.55 ± 0.11	10.32 ± 0.20	93.12 ± 0.09
UNSW	10	Head/CAL64	96.65 ± 0.31	15.41 ± 0.89	90.62 ± 0.30
UNSW	20	Frozen/CAL256	97.15 ± 0.09	19.65 ± 0.29	88.75 ± 0.15
UNSW	20	Head/CAL64	98.13 ± 0.07	25.35 ± 1.37	86.39 ± 0.70

4.3 Calibration targets and precision–recall

Table 5 evaluates common calibration FPR targets. These controllers supply a causal check independent of the $\lambda = 1$ objective. Their realized FPRs must be examined before comparing recall: the same calibration target can produce different future false-alarm burdens. In particular, a small CAL64 sample and a changed score can miss the requested target substantially.

At the nominal 20% target on TON, Head/CAL64 has 91.50% recall versus 80.53% for Frozen/CAL256, a difference of 10.97 points. Their achieved FPRs are 19.95% and 18.37%, respectively: this supports a useful comparison in a nearby operating region, not an exactly matched-FPR estimate. At targets 5% and 10%, the head instead produces FPRs of 12.07% and 16.03%. With CAL-role256 those rates remain 11.21% and 14.49%; larger memory improves these violations but does not remove them. The small paired interpolation difference near 13% FPR supports operating-region dependence, not statistical equivalence or absence of benefit at every lower FPR. In fact, the earlier five-weight interpolation at 10% uses only four supported pairs and is more variable. Target-based policies and cost-weight interpolation describe different sets of operating policies.

The $\lambda = 1$ contrast alone cannot diagnose sampling noise as the cause. On TON, weighting valid calibration/next-block pairs by the next block's normal count gives calibration FPR 42.67% and subsequent FPR 48.30% at $\lambda = 1$, versus 8.26% and 13.20% at $\lambda = 2$. Thus the selected calibration policy already accepts many false alarms; the subsequent rates also reflect transfer. The diagnostic supplement compares calibration and subsequent-block errors and records the number of distinct normal scores. These measurements separate a policy that accepts many false alarms in its calibration objective from one that violates an intended FPR target after transfer. They do not identify a unique causal source of the observed shift.

Figure 2 reports precision–recall at the actual causal operating points. TON's attack-dominated prevalence can yield high precision despite a large FPR. To make that dependence visible, the third panel uses the same class-conditional rates at hypothetical 1% prevalence:

$$\text{precision}(\pi) = \frac{\pi \,\text{recall}}{\pi \,\text{recall} + (1 - \pi)\,\text{FPR}}. \tag{3}$$

For example, the TON head at $\lambda = 1$ yields 4.37% precision at 1% prevalence and 0.45% at 0.1%. These are conditional calculations, not predictions for a new network.

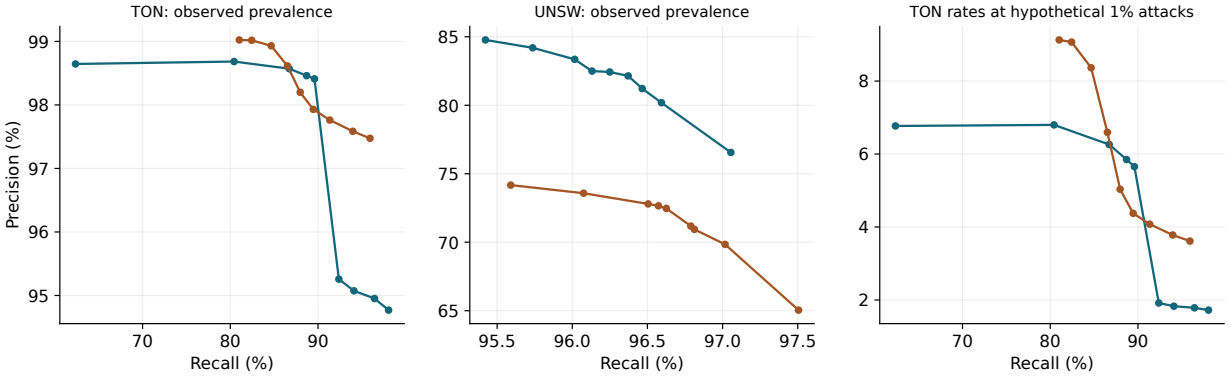


Figure 2: Precision–recall operating points selected by the fixed cost grid. The first two panels use observed class counts. The last panel assumes that TON's class-conditional rates transfer to 1% attack prevalence. Points are seed means conditional on each capture; they are not pooled adaptive-score PR curves.

Table 6: Initial-model audit: pooled AUROC, mean ± seed SD conditional on each capture. A describes the training population and is not an out-of-sample estimate. The chronological and shuffled forests use the same sampled A rows.

Stream	Initial model	A	B	D
TON	Spline	0.998 ± 0.000	0.798 ± 0.003	0.821 ± 0.020
TON	LR	0.884 ± 0.001	0.400 ± 0.003	0.318 ± 0.007
TON	ARF, chronological A	0.703 ± 0.196	0.826 ± 0.226	0.425 ± 0.023
TON	ARF, shuffled A	0.999 ± 0.000	0.986 ± 0.003	0.776 ± 0.108
UNSW	Spline	0.992 ± 0.000	0.977 ± 0.000	0.973 ± 0.000
UNSW	LR	0.981 ± 0.001	0.933 ± 0.052	0.922 ± 0.066
UNSW	ARF, chronological A	0.537 ± 0.257	0.441 ± 0.288	0.439 ± 0.278
UNSW	ARF, shuffled A	0.994 ± 0.001	0.993 ± 0.002	0.993 ± 0.002

4.4 Baseline quality changes the interpretation of learning

The frozen LR on TON has pooled D AUROC 0.318 and Frozen/B BA 25.50%. We therefore audited attack-label orientation, the coefficient score, `decision_function`, attack probabilities, optimization convergence, and the cached score. The fixed-score A/B/D diagnostics are in Table 6. An inversion on B or D is an observed transfer failure; it is not repaired by reversing scores using D labels. Consequently, the poor frozen LR is retained as a failure case rather than presented as a strong performance comparator.

Against its own Frozen/CAL256 control, the TON LR head reduces FP in all five seeds and FN in four. The exception is seed 45, with 62,138 additional FN. This is an exploratory 4/5 joint-improvement count for a different learner and comparator; it does not overturn the failed historical spline criterion. Its interpretation also depends on the frozen LR's poor ranking. Similar head BA values for LR and splines conceal different starting points and error changes.

The initial chronological forest is also weak. On UNSW, seed 43 has a constant score, AUROC 0.5, and recall and FPR of 100% at the historical threshold. Three other seeds have inverted pooled ranking. TON's frozen forest has pooled AUROC 0.40–0.46. Large adaptive-versus-frozen differences against these states cannot alone establish the superiority of a richer learner. Table 7 therefore includes the shuffled-A initialization and the disjoint-label feedback control, while retaining the original chronological results.

For the original chronological initialization, adaptive ARF/CAL256 reaches 90.39% BA on TON, compared with 86.78% for spline Frozen/CAL256 at $\lambda = 2$ and 87.55% for the spline head at $\lambda = 2$. On UNSW it reaches 95.86%, close to spline Frozen/CAL256 at $\lambda = 1$ (95.42%) and below

Table 7: Forest diagnostics, percentages as mean ± SD across five randomizations conditional on each capture. All policies share acquired labels; FIT-only uses a subset for learning and disjoint CAL64. Tree state is persistent and is not constrained to the head's example-memory contract.

Stream	Initial/feedback condition	Policy	Recall	FPR	BA
TON	Chronological A	Frozen/CAL256	76.32 ± 15.88	40.57 ± 8.27	67.87 ± 8.19
TON	Chronological A	ARF/CAL64	95.95 ± 1.11	20.92 ± 3.05	87.52 ± 1.59
TON	Chronological A	ARF/CAL256	96.41 ± 0.64	15.62 ± 1.94	90.39 ± 0.97
TON	Shuffled A	Frozen/B	32.24 ± 8.89	16.69 ± 10.39	57.78 ± 5.46
TON	Shuffled A	Frozen/CAL256	82.47 ± 9.06	35.14 ± 6.33	73.67 ± 5.03
TON	Shuffled A	ARF/B	90.61 ± 2.88	1.86 ± 1.35	94.37 ± 1.11
TON	Shuffled A	ARF/0.5	98.02 ± 0.23	17.38 ± 1.51	90.32 ± 0.67
TON	Shuffled A	ARF/CAL64	96.85 ± 0.75	20.11 ± 4.17	88.37 ± 2.09
TON	Shuffled A	ARF/CAL256	96.81 ± 0.37	16.93 ± 1.26	89.94 ± 0.47
TON	FIT-only feedback	ARF/CAL64	95.29 ± 0.63	20.44 ± 2.56	87.43 ± 1.27
UNSW	Chronological A	Frozen/CAL256	47.57 ± 47.74	26.09 ± 42.15	60.74 ± 15.90
UNSW	Chronological A	ARF/CAL64	97.21 ± 0.92	8.53 ± 1.62	94.34 ± 0.86
UNSW	Chronological A	ARF/CAL256	96.48 ± 1.03	4.77 ± 0.76	95.86 ± 0.38
UNSW	Shuffled A	Frozen/B	97.97 ± 1.37	2.29 ± 0.31	97.84 ± 0.54
UNSW	Shuffled A	Frozen/CAL256	98.02 ± 0.81	4.14 ± 0.64	96.94 ± 0.57
UNSW	Shuffled A	ARF/B	94.57 ± 2.03	3.31 ± 1.03	95.63 ± 0.80
UNSW	Shuffled A	ARF/0.5	89.31 ± 2.24	2.21 ± 0.53	93.55 ± 1.20
UNSW	Shuffled A	ARF/CAL64	98.46 ± 0.19	7.17 ± 1.13	95.65 ± 0.57
UNSW	Shuffled A	ARF/CAL256	97.99 ± 0.32	4.50 ± 0.14	96.74 ± 0.21
UNSW	FIT-only feedback	ARF/CAL64	97.05 ± 1.55	8.40 ± 0.88	94.32 ± 0.66

spline Frozen/B (96.29%). These are descriptive comparisons among the evaluated policies at equal acquisition budgets. Their different retained state and effective label use remain part of the result.

On TON, shuffled-A ARF/CAL256 reaches 89.94% BA; its corresponding frozen/CAL256 state reaches 73.67%, and its historical frozen/B policy reaches 57.78%. The chronological FIT-only/CAL64 forest reaches 87.43%, using CAL-role labels exclusively for threshold selection.

On UNSW, shuffled-A ARF/CAL256 reaches 96.74% BA; its corresponding frozen/CAL256 state reaches 96.94%, and its historical frozen/B policy reaches 97.84%. The chronological FIT-only/CAL64 forest reaches 94.32%, using CAL-role labels exclusively for threshold selection.

At 20% FPR, within-seed interpolation of the shuffled forests' nine-weight policies gives adaptive-minus-frozen recall 18.24 ± 13.88 percentage points across all five supported pairs. This comparison uses adaptive CAL64 versus frozen CAL256, and the forest retains persistent state and calibration-label reuse. It supports a TON learning contribution under that contract beyond the spline-specific single-point contrast. The selected matched-FPR targets have no common support for all five UNSW pairs; no extrapolated forest comparison is used there.

These initialization and label-use contrasts delimit the forest result. The shuffled initial state is a batch-initialization control on historical A, not a reshuffling of the evaluation stream. The FIT-only control removes calibration-label reuse during D while preserving the forest's accumulated tree state; it supplies a label-use check rather than an equal-memory comparison.

Figure 3 places the forest comparisons in their operating regions. The historical-threshold point can outperform recent calibration on stable ranking, while an adaptive policy can help after poor transfer. A comparison at one weight, including an ARF comparison, remains a statement about the specified complete policies. The Supplement's CSV includes paired matched-error descriptions for these forests as well.

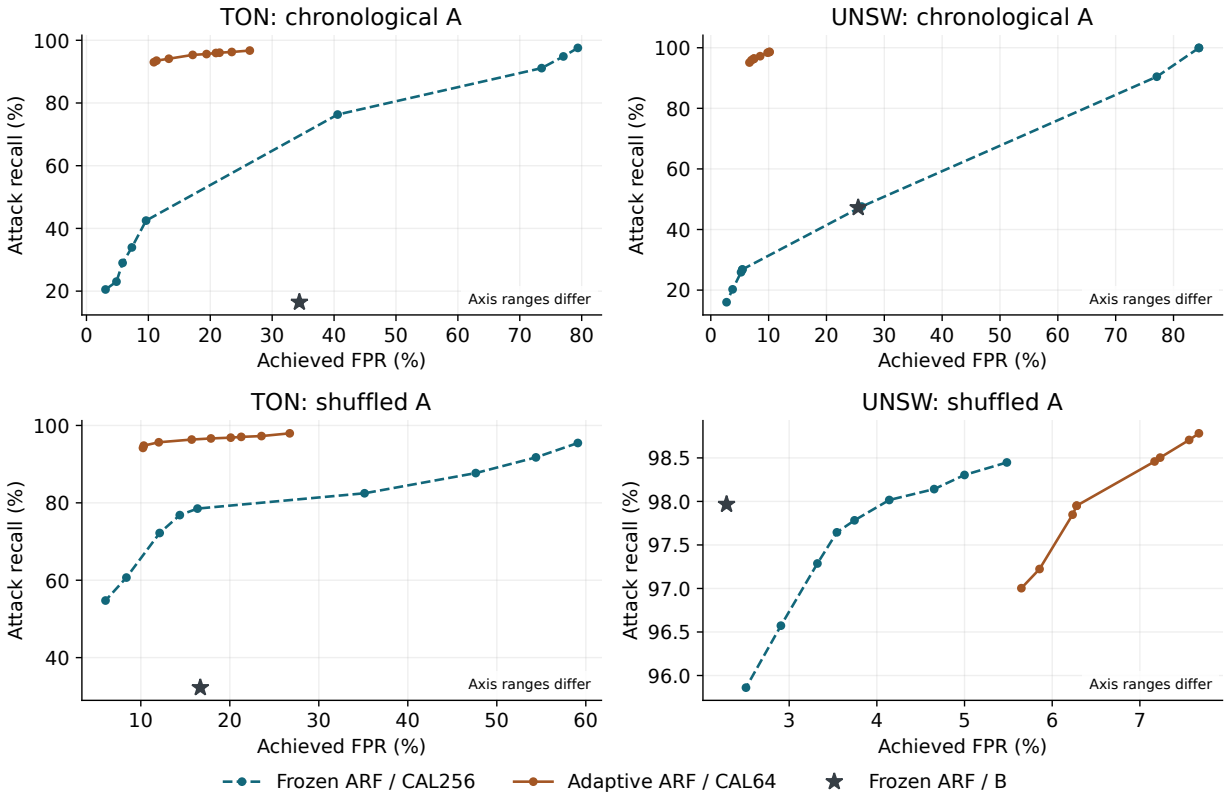


Figure 3: Forest operating policies under chronological and shuffled historical initialization. Connected points use the nine fixed cost weights, with thresholds selected from past labels. Stars mark the fixed historical B threshold. Means describe five randomizations on each stream; axis ranges differ. The frozen control uses CAL256 and the adaptive forest CAL64; the standard forest reuses CAL labels for learning and retains persistent tree state.

4.5 Ranking heterogeneity and controller sensitivity

The frozen spline's TON macro block AUROC is 0.850, while weighting valid blocks by their normal counts gives 0.557. Blocks with few normal observations have disproportionate influence on the macro summary relative to their contribution to false-alarm workload. The normal-weighted statistic exposes weak average within-block separation under that weighting. It does not imply that every normal-heavy block is near chance, nor does AUROC alone determine FPR at a selected threshold. Both summaries, pooled fixed-score AUROC, blockwise AP, and coverage are retained in Supplement S4.

Controller robustness is assessed one factor at a time on both captures: FIT/CAL capacities 96/32 and 384/128; calibration ages 10 and 100 blocks; a minimum of ten CAL examples per class; and two additional fixed role salts. All five seeds are retained for every setting. These checks assess the declared controller rather than selecting a new best configuration. They complement the earlier TON delay and budget tests: at a 0.1% query budget the default spline head reaches 57.49% BA, compared with 79.48% for Frozen/CAL256. Detailed rates, update counts, and conditional variation are reported in the Supplement.

On TON, head BA across the eight declared settings ranges from 81.49% to 88.97%. The default and the two alternative role salts yield 84.84%, 83.95%, and 88.97%, respectively. On UNSW, head BA across the eight declared settings ranges from 89.63% to 93.89%. The default and the two alternative role salts yield 93.22%, 92.44%, and 89.63%, respectively. The default TON seed SD is only 0.79 BA points. It conditions on one chosen role salt and memory configuration, whereas the 81.49–88.97% range changes those choices. Neither quantity is a confidence interval for performance

on a new stream. The three fixed salts give separate TON means of 84.84%, 83.95%, and 88.97%; their within-salt SDs are 0.79, 1.50, and 1.41 points. Supplement S6 reports all salts separately. Three deterministic partitions do not estimate a population distribution over arbitrary salts, and the one-factor checks do not estimate joint configuration uncertainty. The observed salt dependence is substantial and differs between captures. A favorable individual partition should therefore not be reported as a stable model advantage. Across these eight head settings, UNSW BA remains below the historical frozen spline.

4.6 Historical Stage 1 and its numerical sensitivity

The archived TON campaign established the original motivation. With a frozen score, recent CAL64 raises recall from 21.11% to 89.09%, while FPR rises from 12.64% to 46.74%. Adding head refitting gives recall 89.38% and FPR 20.47%. Of the FP reduction, 96.7% occurs on vectors previously observed in A/B/C, where recall decreases. The full-stream FN mean combines opposing seen/unseen changes. Only seeds 43, 45, and 46 improve both error counts, so the specified 4/5 criterion fails.

The same-query Linux numerical bridge changes that count from 3/5 to 2/5. Its paired FN differences are +33,457, −115,004, +123,984, −45,794, and +47,576 for seeds 42–46, respectively. The seed-46 change is largely associated with its frozen calibration trace. This sensitivity is evidence against treating a seed-count cutoff as an inferential conclusion. The original measurements remain historical evidence; the bridge is a separately fitted reconstruction, not bitwise replication. Full counts and provenance appear in Supplement S2–S3.

5 Discussion and limits

The complete-policy comparison and the matched-memory audit answer different questions. The former compares usable policy packages under a common acquisition budget; the latter isolates the effect of changing the score under identical threshold evidence and timing. Equal λ comparisons are meaningful tests of two specified calibration procedures, but they cannot establish uniform superiority of their score functions. Matched-error descriptions and causal FPR-target controllers answer complementary questions. A strong evaluation should report the operating region, actual achieved rates, effective calibration evidence, and the quality of the frozen reference.

The overlap result is similarly conditional. The first extension's $\lambda = 1$ spline comparison locates 90.5% of FP savings on previously seen vectors. This percentage describes where that policy saves false alarms. It is not the fraction of benefit caused by memorization, and it changes when the control or operating objective changes. Exact feature equality can merge distinct flows and miss near-duplicates. A tolerance-based or host/session analysis would require a separately justified identity definition; it is outside the present eight-feature recurrence claim.

The two captures support controlled case-study conclusions. They do not estimate effectiveness across operational networks. Seed SDs, block summaries, and historical moving-block resampling all condition on the recorded streams. Seed SDs additionally fix the role partition and controller configuration, and therefore omit the substantial between-configuration sensitivity shown here. The original D structural checks and earlier project exposure are disclosed; new protocols and runner hashes are fixed before their respective diagnostic executions. Holding D labels out of controller decisions prevents look-ahead within replay, while the broader exploratory research history remains.

Feedback is also an assumption. At the 1% budget, TON requires approximately 566 labels per recorded hour and UNSW approximately 1,288. Block-delay medians are about 3.50 and 6.45 minutes, respectively. These volumes and delays describe an annotation service, not a demonstrated analyst workflow. Labels are treated as correct. Active acquisition, noisy or asynchronous analyst

feedback, and operational capacity estimation require new experiments. A deployment-specific λ should be set from past evidence and declared costs or capacity, with an evaluation period kept separate.

Reproducibility has two levels. Original Windows reports are preserved and reanalyzed exactly, while the available Linux reconstruction does not reproduce every initial fit or prediction. The new extensions retain initial models, source hashes, input-array digests, full per-block reports, and pinned dependencies so that their recorded analyses and causal replays can be checked. A Git revision identifies the implementation; long-term archival deposition is a separate obligation.

6 Conclusions

The apparent value of adaptive intrusion detection depends on the complete decision policy. Matched calibration-memory controls show a persistent TON score-adaptation trade-off, so the original FP reduction cannot be dismissed as a capacity difference alone. Stronger all-label threshold-only controls substantially narrow that advantage in some operating regions. A nominal 20% FPR target yields about eleven additional recall points for Head/CAL64 at nearby achieved FPRs, whereas the paired difference near 13% is small and variable. Stricter requested FPR limits are violated by the tested head controllers, including the larger CAL-role memory.

A practical finding shared by the captures is that the highest-BA evaluated policies retain the historical B threshold: adaptive shuffled ARF/B reaches 94.37% on TON, while frozen shuffled ARF/B reaches 97.84% on UNSW; the historical frozen spline on UNSW reaches 96.29%. The TON forest still learns its score and uses a more permissive state and label-use contract. Thus these observations motivate testing a fixed historical threshold before replacing it with a small recent calibration sample. They do not establish a universally optimal historical policy: the best values are identified retrospectively from these captures, and TON's historical frozen spline is weak.

The contribution is an auditable evaluation procedure and an exploratory two-capture case study. Its essential controls match calibration observations and timing, distinguish capacity from label eligibility, inspect achieved error rates across operating regions, audit initial scores, and report partition/configuration sensitivity separately from conditional seed variation. This supports specific error-trade-off claims and a concrete procedure for testing adaptation, without relying on architectural novelty.

Data and code availability

TON_IoT and UNSW-NB15 are used under their providers' academic-use conditions [27, 28]; raw network records are not redistributed. The package provides source attribution, input and parser hashes, reconstruction instructions, fitted initial states, and full per-block reports. Historical evidence is retained at `81d99d9`, first-extension results at `54871f0`, earlier diagnostics at `dd9e0ea8`, and the matched-memory code and results at `c2401824`. The revision branch and delivery manifest identify the article sources and final revision. Building the paper from recorded evidence requires no raw data; replaying predictions is a separate documented workflow. A permanent public archival identifier has not yet been registered. Deposit metadata and the complete versioned package are supplied; durable archival availability remains pending.

Author contributions and declarations

CRediT authorship contribution statement. Oleksandr Kuznetsov: Conceptualization, Methodology, Formal analysis, Investigation, Software, Validation, Visualization, Supervision,

Project administration, Writing—original draft, Writing—review and editing. Emanuele Pio De Bernardis: Software, Investigation, Data curation, Validation, Visualization, Writing—original draft, Writing—review and editing. Emanuele Frontoni: Funding acquisition, Project administration. De Bernardis developed the original implementation and experimental campaign. Kuznetsov directed the additional diagnostic protocols, computational analyses, and manuscript revision; the computational assistance used for the extensions is described in Methods.

Funding and acknowledgements. GAIA S.r.l., a spin-off of the University of Macerata, supported Emanuele Pio De Bernardis's research through a research collaboration covering August–December 2026. The authors acknowledge this support for the continuation and consolidation of the research programme.

Declaration of competing interests. Emanuele Pio De Bernardis has a funded research collaboration with GAIA S.r.l. Emanuele Frontoni serves on the board of directors of GAIA S.r.l. Oleksandr Kuznetsov supervises the supported collaboration and is involved in a separate research-consultancy agreement between eCampus University and GAIA S.r.l. These relationships are disclosed for transparency.

Ethics and data use. The work uses existing research datasets, without new participant recruitment or live-network attack deployment.

Declaration of generative AI and AI-assisted technologies in manuscript preparation

During preparation of this work, the authors used OpenAI ChatGPT/Codex to assist with literature organization, text structure, scientific editing, and consistency checks. Its use for experimental code, numerical analysis, and figures is described separately in Methods. The human authors retain responsibility for the scientific claims, references, authorship, and content of the submitted work.

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